J Oral Maxillofac Surg
55:275-281, 1997

Management of Dental Extractions in Irradiated Jaws: A Protocol Without Hyperbaric Oxygen Therapy

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The absolute incidence and prevalence of osteoradionecrosis (ORN) of the jaws after radiation therapy (RT) for treatment of oral and oropharyngeal cancer are unknown. Since 1922, when Coutard[1] reported the Fondation Curie's experience with ORN, one finds that it has been observed to occur in both dentulous and edentulous patients, either spontaneously or after wounding. In particular, ORN has been noted to occur after dental extractions performed shortly before or at any time after RT. Reviewing studies from the literature, most of which are retrospective, one finds an overall incidence of ORN of 11.8% before 1968 (Table 1) and 5.4% after that date (Table 2), by which time almost all radiation oncology units had embraced megavoltage or supervoltage therapy. These data were based only on the presence of ORN without subcategorization into dentulous, edentulous, spontaneous, or extraction related. Unfortunately, this risk, although highest during the first 4 to 12 months after RT, has been found to persist for the remainder of the patient's life.[1-10]

The older concept of ORN developing consequent to a triad of RT above a critical dose, local trauma, and infection was articulated in 1938 by Watson and Scarborough[11] with support by others.[12,13] In the 1980s this concept was challenged by Marx,[14,15] who suggested an alternate hypothesis that ORN resulted from radiation-induced, deficient cellular turnover and collagen synthesis in an hypoxic, hypovascular, and hypocellular environment in which tissue breakdown exceeded the repair capabilities of the wounded tissue.[16]

Because the presence of carious and periodontally compromised teeth in the irradiated mandible has long been associated with ORN, controversy has existed regarding whether such teeth should be removed before or after RT, whether they should ever be removed, and, if they should be removed, when, by what technique, and by whom. Thirty years ago it was recognized that the maintenance of a healthy dentition was essential for the prevention of ORN. This required the willing participation of a dentist to assess and monitor the dentition and to remove teeth as necessary.[16,17] All teeth that were grossly carious, periodontially hopeless, or had a poor prognosis for retention beyond 12 months[8] were removed before RT,[16] with extractions after RT being avoided as much as possible.[8] Fully embedded teeth were not removed.[16,18] At the time of extraction, alveoloplasty with primary closure, the use of preoperative and postoperative antibiotics, and a 10-day wait before starting RT were recommended. Although the use of systemic antibiotics was well supported in the literature of this time,[7,17,19-21] a more recent article by Marciani and Ownboy[22] reported excellent results both for pre-RT and post-RT extractions without using antibiotics. After the adoption of an organized plan to manage the dentition, and the use of high-energy radiation therapy in the years after 1968, the rate of ORN for extractions performed before RT fell to 4.4% (Table 3).

Most studies have reported a higher rate of ORN after dental extractions after RT than before it. However, in carefully evaluated patients who received aggressive preventive dental care and whose extractions were performed by experienced oral and maxillofacial surgeons, the five most recent studies have reported a prevalence of ORN of only 1.1% in 424 patients who had teeth removed after RT.[21-25]

Most authors recommend alveoloplasty and primary closure followed by a 10- to 14-day wait before starting RT,[7,16-18,23,25-29] although a minority recommend waiting 21 days.[30,31] For postradiation extractions, there has been more conservatism in the recommendations, with advice including simple forceps extraction with due regard to not damaging underlying bone,[16,20,26] minimal

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0278-2391/97/5503-0012$3.00/0

Table 1. Incidence of Postextraction Osteoradionecrosis (ORN)

No. of Patients Irradiated	No. of Cases of ORN	First Author	Year Published
1,197	213	Watson[11]	1938
103	26	Martin[53]	1940
104	6	Wildermuth[28]	1953
491	26	Meyer[12]	1958
143	10	Friedman[55]	1959
108	10	Dodson[56]	1962
24	16	MacComb[8]	1962
593	60	Hayward[16]	1963
364	18	MacDougall[57]	1963
185	6	Shimanovskaya[58]	1964
3,312	391 (11.8%)		

mucoperiosteal flap elevation,[17,22,26] minor alveoloplasty and primary closure,[26,29] finger extraction of loose teeth, and waiting until spontaneous exfoliation occurs.[17,19]

All authors agree with Beumer et al[32] that ORN is most likely to develop after removing a mandibular molar located directly in the treatment field. In general, the risk for the development of ORN is found to be significantly higher for dentulous than for edentulous patients[23,24,31,33]

To evaluate whether there is a place for adjunctive hyperbaric oxygen (HBO) therapy before extracting teeth in irradiated mandibles, one must first consider the natural history of ORN. This begins with defining the nature of ORN, determining its prevalence, and then considering the consequences of acquiring ORN. Lastly, one would like to know what is the probability that ORN could be prevented by prophylactic HBO therapy in irradiated patients who require dental extractions and what would be the cost of its use.

Table 2. Incidence of Postextraction Osteoradionecrosis (ORN) After 1968

No. of Patients Irradiated	No. of ORN	First Author	Year of Publication
593	6	Hayward[16]	1968
31	0	Solomon[29]	1968
176	12	Rankow[42]	1971
278	10	Beumer[59]	1972
49	2	Carl[60]	1972
304	66	Daly[27]	1972
74	13	Cheng[61]	1974
220	23	Marciani[62]	1974
109	3	Marciani[22]	1986
23	1	Ange[63]	1975
381	54	Bedwinek[35]	1976
88	6	Beumer[64]	1976
404	32	Murray[33]	1980
528	11	Horiot[65]	1981
100	22	Morrish[36]	1981
2,853	22	Coffin[17]	1983
1,989	270	Fleming[19]	1983
94	0	Makkonen[24]	1987
431	21	Widmark[26]	1987
72	0	Maximiw[25]	1991
1,140	14	Kumar[7]	1992
11,077	602 (5.4%)		

Table 3. Incidence of Osteoradionecrosis After Preradiation Dental Extractions Reported After 1968

Study/Year	No. of Patients With preRT Extractions	No. of Cases ORN (%)
1. Kumar '92[7]	728	4 (0.5)
2. Widmark '89[26]	102	4 (4.0)
3. Makkonen '87[24]	45	0 (0)
4. Marciani '86[22]	109	3 (2.8)
5. Beumer '83[32]	120	17 (14.2)
6. Horiot '81[65]	528	11 (2.1)
7. Morrish '80[36]	22	3 (13.6)
8. Starcke '77[40]	62	1 (1.6)
9. Bedwinek '76[35]	203	24 (11.8)
10. Beumer '76[64]	17	3 (17.6)
11. Daly & Drane '72[27]	74	22 (29.7)
	178	4 (2.2)
Totals	2,188	96 (4.4%)

Definition

ORN is a nonhealing, nonseptic lesion of bone in which bone volume and density cannot be maintained by the hypocellular, hypovascular, hypoxic tissue, which cannot adequately meet its metabolic demands.[15] This pattern of ORN (type I) may present with bone lysis under intact gingiva or mucosa. When the soft tissue breaks down, the bone becomes exposed to saliva and, secondary contamination occurs. In those cases in which sepsis is directly introduced by the trauma of dental extraction or surgery, an aggressive form of ORN (type II) may develop, which is better described as a radiation osteomyelitis.[34] Type I ORN tends to heal with conservative therapy and the type II does not, but most of the literature does not differentiate between the two nosologically, categorizing them both simply as ORN. Approximately half of all cases of ORN reviewed in this article healed with conservative treatment (Table 4).

Occurrence

Since the introduction of higher-energy RT in the mid to late 1960s, the incidence of ORN has fallen

Table 4. Results of Conservative Treatment of Mandibular ORN

Author	ORN	Healed	Healed and Stable
VAMC '95[54]	3	3	3
vanMerkestyn '95[47]	16	2	2
Barrelier '91[67]	11	7	7
Widmark '89[26]	21		7
Marciani '87[22]	3	3	3
Kluth '87[68]	6	5	5
Beumer '83[32]	13		8
Morrish '80[36]	16	4	14
Murray (I) '80[33]	44	19	24
Murray (II) '80[69]	57	36	42
Murray (III) '80[70]	24		21
Bedwinek '76[35]	54		37
Beumer '76[64]	3	3	3
Ange '75[63]	4	4	4
Daly '72[27]	66		20
Rankow '71[42]	21		9
Cachin '69[2]	22		16
Grant '66[71]	69		50
MacComb '62[8]	32	25	25
Total	476	111/189 (58.7%)	300 (63.0%)

Reports including >50 patients: Complete healing

	Healed	ORN
Murray[33]	36	57
MacComb[8]	54	93
Daily '72[27]	20	66
Totals	110 (50.9%)	216

from 11.8% (Table I) to 5.4% (Table 2). However, spontaneous, dentulous versus edentulous, denture trauma–induced, surgical trauma–induced, and preradiation and postradiation extraction–induced ORN are lumped together in the statistics. Therefore, separation by type is only possible for limited samples. Consequently, the best contemporary estimate of ORN developing at any time after RT is 5.4% (Table 2). Superimposed on this baseline rate is the chance of developing ORN after dental extractions, which ranges from 0 to 65%, but averages 5.8% when the data from different studies are pooled (Table 5). These data also combine mandibular and maxillary ORN; but the contribution of maxillary ORN is negligible because its occurrence is uncommon. Moreover, there are inadequate data to factor out the baseline rate of spontaneous ORN to give an accurate value for the postextraction incidence of ORN arising independently from the spontaneous "background" in dentulous patients.

Radiation Dosimetry

An additional confounding variable in data analysis is the relationship between radiation dose and ORN. Animal and human studies have both related the probability of acquiring ORN to dose. At treatment levels below 50 Gy ORN is uncommon, but not impossible. Above 60 Gy, but below 70 Gy, ORN occurs more frequently, and above 75 Gy the rate is almost 10 times greater than it is for less than 50 Gy.[5,7,35,36] In addition, the chance of healing of ORN lesions after conservative treatment diminishes significantly for cases induced by doses above 75 Gy.

Conservative Therapy

Of the patients who develop ORN, those who do so spontaneously tend to heal after conservative therapy.[27,31] This is especially true for those patients who are edentulous and whose ORN was induced by a den-

Table 5. Incidence of Osteoradionecrosis (ORN) After Postradiation Dental Extractions Reported Since 1968

Author/Year	Patients	Extracted Teeth Within RT Treatment Area	Teeth Extracted in Patients Receiving RT	ORN (%)
1. VAMC '95[54]	100	—	—	3 (3)
2. Maxymiw '91[25]	72	196	449	0 (0)
3. Widmark '89[26]	19	50	50	2 (4)
4. Makkonen '87[24]	25	—	94	0 (0)
5. Marciani '86[22]	17	—	109	0 (0)
6. Marx '85[52]	37	—	—	11 (30)
7. Beumer '83[23]	—	—	72	16
8. Morrish '80[36]	18	—	—	4 (2)
9. Bedwinek '76[35]	178	—	—	4 (2)
10. Beumer '76[64]	11	—	>99	1 (9)
11. Carl '73[66]	47	187	187	2 (4)
12. Solomon '68[30]	31	—	—	0 (0)

NOTE. Last decade (1-5) nORN/nRTX = 5/233 = 2.1%; Overall: (1-6, 8-12) = 32/555 = 5.8%. 1) Not all articles separated maxillary from mandibular ORN. 2) Number of extractions were not separated into maxillary and mandibular in most studies. 3) Study 6 reported 72 "extraction episodes" not otherwise specified. Number of patients not reported.

ture irritation. Denture irritation or ulceration by itself was reported as being an inadequate stress to induce ORN.[68] In dentulous mouths, Kumar et al[7] found that most of his patients who developed ORN did so at sites where teeth had been extracted in the recent but not immediate past, not as part of planned preradiation therapy extractions. None of these lesions progressed to pathologic fracture or loss of continuity.

When ORN presents as a nonhealing, stable or slowly progressive ulceration of the mucosa with exposed underlying bone, conservative treatment consisting of systemic antibiotics, topical antibiotics or antiseptics, and oral irrigations has been recommended.[2,17,21,35,38,39] Such conservative treatment, which may need to be continued for years in stable cases,[17] resulted in healing in 56.9% of cases and stabilization or healing in 61.8% (Table 4). However, as with all literature reviews, all data in the articles were not presented in the same way and, therefore, it is not possible to determine the length of long-term assessment in every study. Consequently, a definitive statement regarding the long-term durability of healing after conservative treatment cannot be made. However, Murray et al[33] in their long-term, review of 128 cases of ORN treated conservatively, assigned a constant probability of .06 per month for the termination of the ORN.

ORN and Tooth Removal

The role of dental extraction in relation to ORN merits serious consideration. Extractions have been blamed more often than any other factor for causing ORN. The question of risk in regard to inducing ORN in patients having extractions performed before or after RT is well illustrated by Tables 3 and 5. Although specific reviews have reported an unacceptably high incidence of ORN after both preradiation therapy extractions[27,28,31,32,35,40] and postradiation therapy extractions,[23,35,36,60] reviewing the entire literature since the advent of the modern age of RT, in which megavoltage and supervoltage photons and electrons replaced orthovoltage beams in the mid to late 1960s, one finds that there is little difference between the rate of ORN associated with dental extractions before (4.4%) and after (5.8%) RT. These rate differences may be statistically significant, comparing an at-risk group of 2,188 in the preradiation therapy group with 1,713 in the postradiation therapy group. However, from a clinical perspective, the difference is only two events per 100 patients.

Importantly, it has been noted that the role of senior dentists expert in caring for the dentition of head and neck cancer patients, who could competently remove teeth, was critical to minimizing ORN.[16,17,24-26] What number of teeth per quadrant should be removed at a single setting, whether the periosteum should be elevated, whether sockets should be closed primarily, whether radical or conservative alveoloplasty should be performed, and whether antibiotics should be used, have all been scrutinized in an attempt to reduce ORN after extractions. After 60 years, the issue remains unsettled, but all authors agree that the extraction technique should be as gentle as possible. The reviews from the last decade have reported an ORN rate of 2.1% (5 cases of 233 patients having had dental extractions after RT). For the entire period since 1968, the rate was 5.8% (32 ORN in 555 patients having post-RT extractions). Whether these results were due to a specific extraction technique, avoidance of vasoconstrictor in the local anesthetic, using agents other than lidocaine, or using specific antibiotics remains unresolved. As always, follow-up review varied greatly, ranging from 2 months to 19 years.

HBO, Extractions and ORN

In a continuing search for methods to reduce ORN after dental extraction, the idea arose to use HBO to prevent ORN. Because HBO has been useful in treating established ORN, might it have a role in preventing ORN? To answer this critical question, one must first review the success with HBO in treating ORN. Since the 1970s, HBO has been enthusiastically promoted for treatment of ORN of the mandible. Hart and Mainous.[5,41] were able to eliminate symptoms for 80% (37 of 46) of their patients with severe ORN, while improving symptoms in another 3 of 46. Eleven percent (5 of 46) needed a second series of dives beyond the initial 60. All 46 of these cases were severe enough to meet Rankow and Weissman's criteria for hemimandibulectomy.[42] Farmer et al[43] in 1965 reported on 13 cases of ORN with complete healing in 10 of 13 cases, but three later relapsed, resulting in stable healing in 7 of 13. However, patients were only followed from 11 to 27 months. Although 3 of 13 showed no improvement, four of five with pathologic fractures healed with a fibrous union. Therefore, although many dives were required, very good results were obtained in these severe cases of ORN. Marx and Ames[44] refined the HBO treatment to 20 preoperative and 10 postoperative dives and claimed a 91.6% success rate, which they noted was highly favorable compared with the 20% to 50% success rates reported without HBO.[34,45] Others[21,46] have established clinical and theoretical reasons for the success of HBO and suggested its use before surgical wounding in irradiated patients. A bit less optimistic than Marx was vanMerkesteyn et al,[47] who found that in severe ORN, HBO led to initial resolution of only 15 of 29 of his cases, but a second course of HBO improved the result to 20 of 29 cases (69.0%). Thirteen of these patients had mandibular discontinuities at the start of treatment, and two healed. The total number

Table 6. ORN Resulting in Loss of Continuity of Mandible

Author	Discontinuity/ORN	Percentage Losing Continuity
van Merkestyn '95[47]	5/16	31
Epstein '87[4]		23
Beumer '83[23]		29.5
Beumer '84[38]		29.5
Bedwinek '76[35]	17/54	31.5
Daly '72[27]	46/66	70
MacComb '62[8]	16/16	100
Carl '72[60]	2/2	100
Cheng & Weng '74[61]	13/13	100
Morrish '80[36]	6/22	27.3
Totals	105/189	55.5

of patients who lost continuity was 17 of 29, with 5 of 29 becoming discontinuous during HBO treatment. Disconcerting regarding the protective effects of HBO against later development of ORN was the observation that in two patients a second instance of ORN developed spontaneously on the side contralateral to the index ORN, which then resulted in loss of mandibular continuity on the second side. In vanMerkesteyn's series, HBO dives were of 90 minutes' duration at 3 ATA for 30 preoperative dives and 10 dives after sequestrectomy, debridement, or resection. His patients had received between 60 and 90 Gy (average = 69.6 Gy) and were followed for 3 to 84 months (mean = 16 months).

Three enthusiastic reports on the use of the HBO for treating ORN report loss of continuity to occur from 16% to 31% of the time during HBO treatment.[4,8,47,48] In 37 cases, 23, or 62%, required resection, despite HBO. This is not very different from the 55.5% loss of continuity after conservative treatment reported in Table 6.

Because HBO has had a salutary affect on established ORN, one might be tempted to consider applying this technology to prevent ORN. However, would it be effective in preventing ORN occurring consequent to wounding after dental extraction? If it were effective, how expensive would it be and would the cost be justified in relation to the benefit of preventing ORN? Several authors[31,45,49-51] have suggested and have supplied successful illustrations of its use before dental extractions in irradiated patients. However, only one study by Marx et al[52] prospectively compared two groups of patients, one having HBO and one having penicillin before their extractions. This study found a significant reduction in ORN from 29.9% to 5.4% in a sample of 137 patients, of whom 11 in the penicillin group and 2 in the HBO group developed ORN. From the perspective of statistical assessment, the findings were significant. It is interesting to note, however, that in the particular group of patients being analyzed, the overall rate of ORN occurring after tooth extraction was 35% (13 of 37), which was very high compared with the 5.8% rate for postradiation therapy extractions shown in Table 5. This implies great success for HBO in this particularly high-risk group of 37 patients. However, applicability from this prospective study to the general population of patients in whom the ORN rate is only 5.8% (Table 5) is uncertain.

Several problems remain regarding the use of HBO to prevent ORN after extractions. First, the incidence over the past 28 years is low (5.8%) and may be even lower in the last decade (2.1%). Second, slightly more than half (56.9%, Table 4) of those patients who develop ORN will heal after conservative treatment. The 2.3% of the total group of irradiated patients who developed ORN and did not heal after conservative treatment might have profited from HBO, but even in Marx's most positive study 2 of 37, or 5.4%, still developed ORN even after HBO. In his small sample, this represented a large improvement over the 11 of 37 (29.7%) who developed ORN without receiving HBO. HBO did not prevent all cases of ORN, and therefore its effectiveness was less than 100%. If one were to consider an effectiveness of approximately 80%, the midpoint of the reported range of 69% to 91%,[5,44,47,49,52] then there should be a reduction of ORN from 5.8% to 1.2%, which would be of clinical significance. If one considers only the most recent five studies in Table 5 as representing results using the best extraction techniques, then the rate would drop from 2.1% to 0.4%, which would not be clinically significant. However, in regard to the most serious consequence of ORN, which is the loss of mandibular continuity, this rate might be reduced from 2.5% to 0.5%, representing a reduction of two cases per 100 cases of ORN. Using this most pessimistic assessment, treating 100 patients with HBO to prevent two cases of loss of continuity would cost approximately 1.5 million dollars (data courtesy Hyperbaric Unit, Henry Ford Hospital, Detroit, MI). Although it would be very desirable to eliminate ORN after extractions, the cost of treating 98 patients who would not have benefited from this treatment to prevent the two who would have benefited, may well be unsupportable. If it were possible to determine which patients were a significantly high risk to benefit from HBO to prevent ORN after tooth extraction, HBO could then be recommended for these patients, but this has not been possible.

One must also examine the spontaneous rate of ORN of 37% to 71% in a mixture of dentulous and edentulous patients. A small subset of spontaneous ORN occurring in dentulous patients gave a rate of 30% (16 of 44).[26,36] In a larger series, Bedwinek et al[35] reviewed 381 dentulous patients and found a constant rate of

spontaneous ORN of 5%. Of their 54 patients with ORN, 19 developed spontaneously and, of the remaining 35, 24 occurred in pre-RT extraction cases and four occurred in post-RT extraction cases. If it were true that 30% of irradiated dentulous patients who subsequently developed ORN did so spontaneously, the true postextraction ORN rate would fall from 5.8% to 4.0%. Using the figures discussed in the preceding paragraph, the rate of loss of continuity would be reduced 30%, from 2.5% to 1.75%. In Bedwinek et al's larger series, the spontaneous ORN rate of 4.5% was actually greater than the post-RT extraction ORN rate of 2.3%.

Conclusion

It can be concluded that the true rate of loss of continuity of mandibles that develop ORN after dental extractions varies greatly between small series, but in aggregate review it is quite low. Even if one were to disallow any contribution of spontaneous ORN to the reported postextraction ORN rate, the loss of continuity rate would still be 2.5%. If one were to treat all of these at-risk mandibles with HBO before extractions, the loss of continuity rate might be reduced to 0.5% if HBO were 80% effective, which may be an overly optimistic assessment. At best, therefore, it might preserve two mandibles while unnecessarily treating 98. Thus, these data do not support the mandatory use of HBO before removing teeth in irradiated mandibles, particularly when one considers that in the most recent reports of ORN after dental extractions[22,24-26,72] the rate was only 2.1%.

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